

Answer to Paper 1

Question	Answer	Question	Answer	Question	Answer
1	D	11	A	21	A
2	A	12	C	22	D
3	B	13	C	23	B
4	D	14	B	24	D
5	D	15	C	25	C
6	C	16	B	26	C
7	A	17	C	27	D
8	C	18	C	28	D
9	A	19	B	29	D
10	A	20	D	30	B

Answers to Paper 2

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 The mass spectrometer made it possible to determine the masses of atoms very accurately, and it became clear that not all atoms of the same element had the same mass.

The accurate masses gave rise to the idea of the isotope:

“Isotopes have the same atomic number, but different mass numbers.”

- (a) (i) Give the symbols showing the nucleon numbers and charges of the following three particles.

Particle	Protons	Neutrons	Electrons
K	6	8	6
L	7	7	10
M	8	7	7

K: ${}^{14}_6\text{K}$

L: ${}^{14}_7\text{L}^{3-}$

M: ${}^{15}_8\text{M}^{+}$

- (ii) Use the Data Booklet to identify which are the usual isotopes of the element concerned.

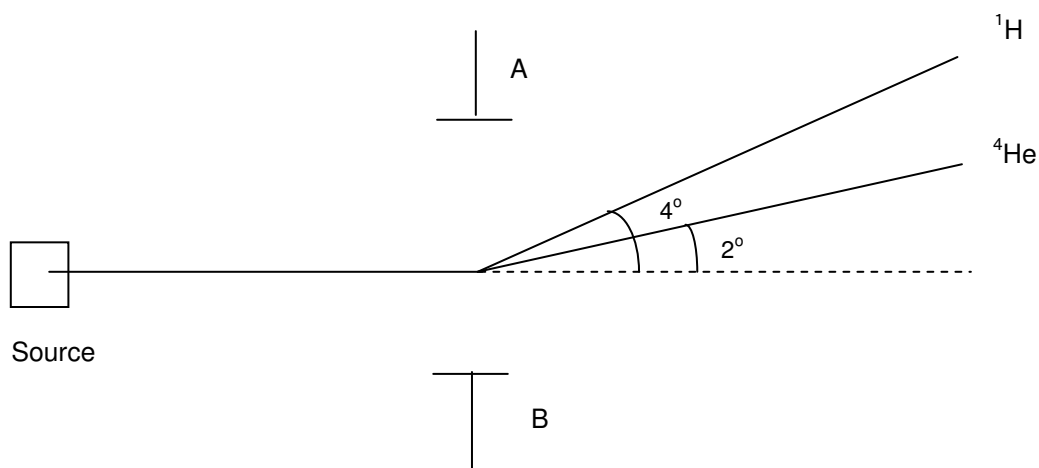
Particle K: ${}^{12}_6\text{C}$

Particle L: ${}^{14}_7\text{N}$

Particle M: ${}^{16}_8\text{O}$

[3]

- (b) A plasma is a gaseous mixture in which the atoms have been completely stripped of their electrons, leaving bare nuclei. Because of possible use in controlled nuclear fusion reactions, plasma behaviour has been intensively studied. When passed between two plates carrying a certain electric charge, ^1H and ^4He nuclei are deflected as follows.



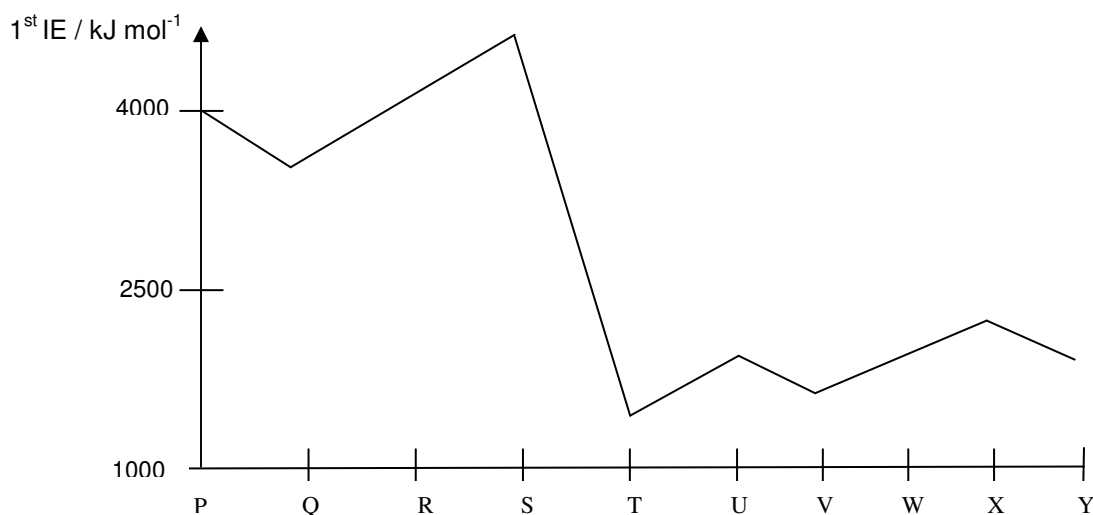
Giving reason for your answer, suggest the polarity (+ or -) of plate A.

Plate A is negatively charged.

As nuclei which are positively charged gets attracted to it.

- (c) In an analytical chemistry laboratory, the plasma torch is often used to ionise a variety of samples during their analyses.

In one series of analyses, samples of unknown elements **P – Y** were passed through the plasma torch and their first ionisation energies were recorded and plotted as shown.



Elements **P – Y** have atomic number below 20.

- (i) Identify the Group in the Periodic Table to which element V belongs and write its ground state electronic configuration.

Element V belongs to Group III

Electronic configuration of V: ns^2p^1

[2]

- (ii) Element X forms more than one stable chloride. State the formula of one of these chlorides. Draw the dot-and-cross diagram of the chloride of X that you have stated and predict its shape.

Formula of the chloride: XCl_3 or XCl_5

Dot-and-cross diagram:



Shape of chloride of X: Trigonal pyramidal or trigonal bipyramidal

[3]

- (iii) There is a pair of elements among the elements **P – Y** that belong to the same group. Identify any pair, stating the group to which they belong.

Pair: **P / Y** Group: **V** [2]

- 2 Surplus alkenes can be used as fuels as well as starting materials for petrochemical processes. Being hydrocarbons, alkenes can burn, transferring energy to their surroundings.

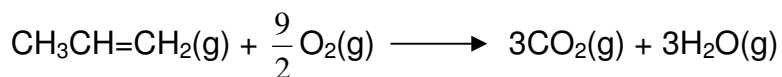
- (a) What do you understand by the term *bond energy*?

Bond energy is the energy required to break 1mol of covalent bonds in gaseous molecules to form gaseous atoms.

[1]

- (b) The equation for a combustion of propene, $CH_3CH=CH_2$, is

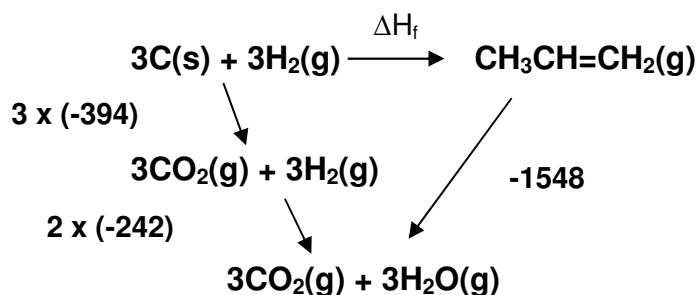
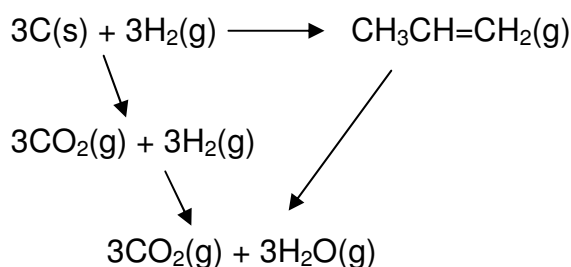
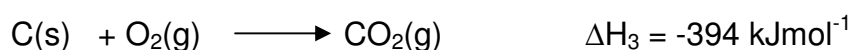
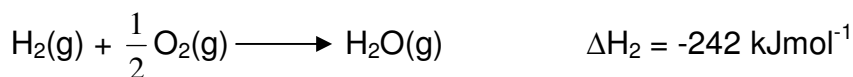




By using appropriate bond energy data given in the *Data Booklet*, calculate a value for the enthalpy change of the above reaction, ΔH_1 , involving propene.

$$\begin{aligned}\Delta H_1 &= \Sigma(\text{E required to break bond}) + \Sigma(\text{E released on bond formation}) \\ &= (+5622) - (7200) \\ &= -1548 \text{ kJ mol}^{-1}\end{aligned}$$

- (c) Using the answer obtained in (b) and with the following enthalpy changes for the reactions given below, use the energy cycle provided to calculate the enthalpy change of formation, ΔH_f , of propene. [3]



$$\begin{aligned}\Delta H_f &= 3(-394) + 3(-242) - (-1548) \\ &= -360 \text{ kJ mol}^{-1}\end{aligned}$$

- (d) A sample of propene was ignited in a calorimeter to heat up 150 cm³ of water. The temperature of the water rose by 45°C.

Calculate the mass of propene that had been combusted.

Assume the specific heat capacity of water to be 4.2 Jg⁻¹K⁻¹.

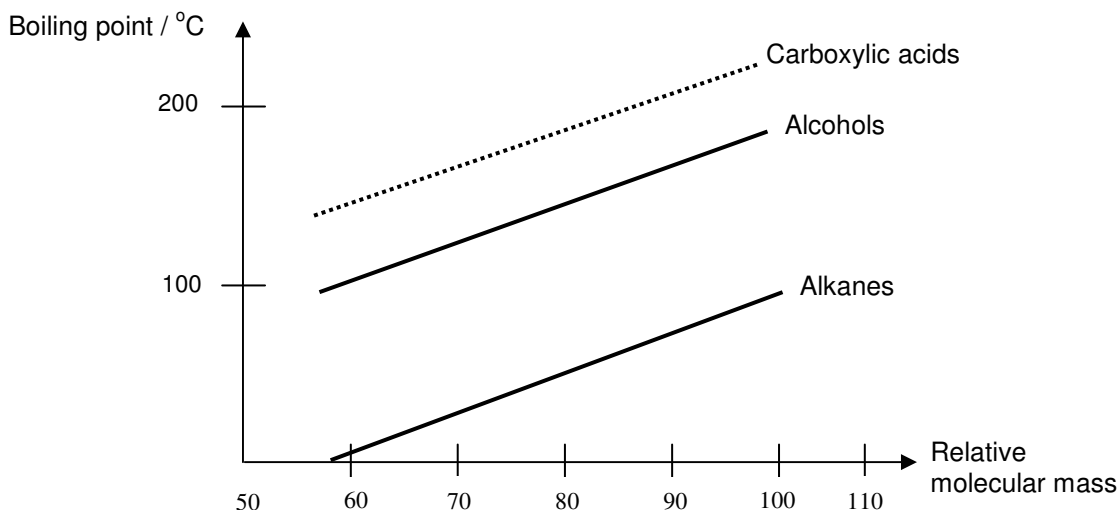
$$q = mc\theta = 150 \times 4.2 \times 45 = 28350 \text{ J}$$

$$n_{\text{propene burnt}} = \frac{28350}{154800} = 0.0183 \text{ mol}$$

$$\text{mass}_{\text{propene}} = 0.0183 \times 42 = 0.769 \text{ g}$$

- (e) Alkanes and alcohols are highly combustible and are very valuable as fuels.

The general trend of boiling points of alkanes and alcohols against relative molecular mass is shown in the graph below.



- (i) Explain, in terms of the bonding present, the trends shown by the above graph.

In both alkanes and alcohols, as Mr increases, IMF (van der Waal's attraction) between molecules get stronger, thus b.p. increases.

Alcohols have a higher b.p. than respective alkanes because of the presence of intermolecular H-bonding.

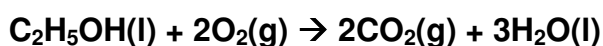
H-bonding is a stronger IMF than van der Waal's.

[3]

- (ii) On the above graph, indicate the trend of boiling point of carboxylic acids against relative molecular mass.
- (f) The table below shows the sources and the enthalpy change of combustion of two common fuels used in vehicles.

Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}
Octane (C_8H_{18})	Refined from crude oil	-5460.6
Ethanol ($\text{C}_2\text{H}_5\text{OH}$)	Hydration of ethane Fermentation	-1366.2

- (i) Define, using octane as an example, the term *enthalpy change of combustion*. Include an equation in your answer.



- (ii) The fuel value (in kJ g^{-1}) of a substance is the heat energy *released* when 1 g of the substance is combusted. Calculate the fuel value for each of the two fuels.

$$\text{Fuel value for octane} = \frac{5460.6}{114} = 47.9 \text{ kJ g}^{-1}$$

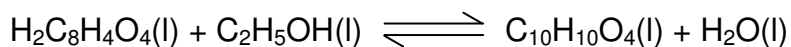
$$\text{Fuel value for ethanol} = \frac{1366.2}{46} = 29.7 \text{ kJ g}^{-1}$$

- (iii) Using the information given in (c) and your answer in (c)(ii), compare the relative advantages and disadvantages of using octane and ethanol as fuels for vehicles.

Octane	Ethanol
Larger fuel value and thus releases more energy per gram of fuel	Smaller fuel value and thus releases less energy per gram of fuel
Requires more sophisticated equipment or tedious process to obtain crude oil and further refine it	Requires relatively simpler equipment or simpler process to manufacture
Fewer sources of obtaining octane	More alternatives / simpler sources of obtaining ethanol

[Total: 17m]

- 3 Upon heating, phthalic acid reacts with ethanol according to the equation as shown below.



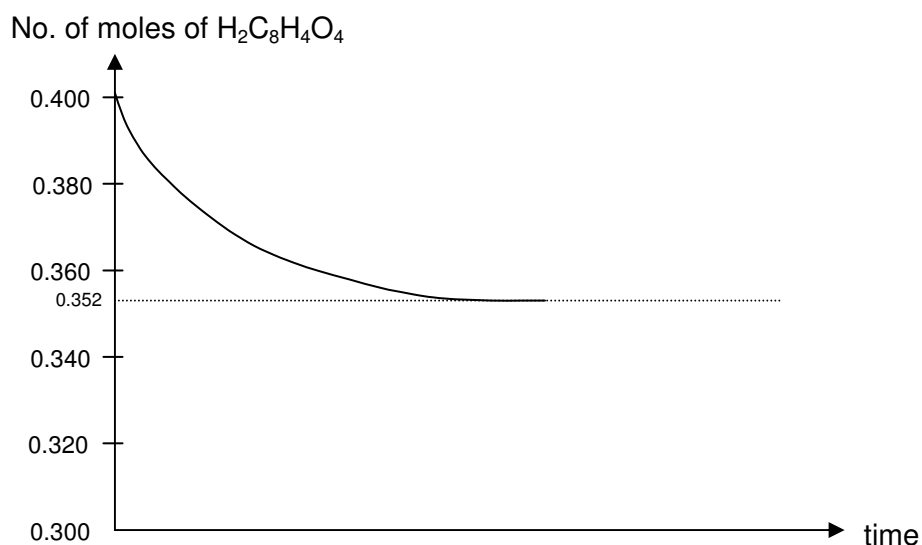
The equation does not really indicate the problems with the reaction. To get a good yield of the product, $\text{C}_{10}\text{H}_{10}\text{O}_4$, the reaction has to be managed carefully since it is reversible.

A mixture with the following composition:

Compound	No. of moles
$\text{H}_2\text{C}_8\text{H}_4\text{O}_2$	0.400
$\text{C}_2\text{H}_5\text{OH}$	0.600
$\text{C}_{10}\text{H}_{10}\text{O}_4$	0.100
H_2O	0.100

was heated in a closed container and the reaction allowed to reach equilibrium.

The graph shows how the number of moles of $\text{H}_2\text{C}_8\text{H}_4\text{O}_2$ varies with time.



- (a) To analyse the composition of the reaction mixture, a fixed volume was extracted at regular time intervals and titrated **rapidly** against a standard solution of alkali. Explain why the titration has to be carried out rapidly.

It's a dynamic system. Reactions are still on-going. Titration to be carried out rapidly to minimize change in equilibrium concentration so that the composition at time t can be analysed accurately.

- (b) Write an expression for the equilibrium constant, K_c , for the reaction above.

$$K_c = [\text{C}_{10}\text{H}_{10}\text{O}_4][\text{H}_2\text{O}] / [\text{H}_2\text{C}_8\text{H}_4\text{O}_4][\text{C}_2\text{H}_5\text{OH}]$$

- (c) What is the value of the equilibrium constant K_c for this reaction. Show how you derive at your answer.

	$\text{H}_2\text{C}_8\text{H}_4\text{O}_4$	$\text{C}_2\text{H}_5\text{OH}$	$\text{C}_{10}\text{H}_{10}\text{O}_4$	H_2O
I	0.400	0.600	0.100	0.100
C	-0.048	-0.048	+0.048	+0.048
E	0.352	0.552	0.148	0.148

$$K_c = 0.113$$

- (d) Explain how the equilibrium composition would change if cold concentrated sulphuric acid was added to the equilibrium mixture?

Addition of concentrated sulphuric acid removes H_2O ; equilibrium position shifts to the right. New equilibrium mixture contains more $\text{C}_{10}\text{H}_{10}\text{O}_4$

- (e) The reaction was repeated at the same temperature with 0.300 moles each of $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$ and $\text{C}_2\text{H}_5\text{OH}$ and 0.200 moles each of $\text{C}_{10}\text{H}_{10}\text{O}_4$ and H_2O . Determine the concentration of $\text{H}_2\text{C}_8\text{H}_4\text{O}_4$ at this equilibrium, in a 1 dm^3 container.

$$[\text{H}_2\text{C}_8\text{H}_4\text{O}_4] = 0.374 \text{ mol dm}^{-3}$$

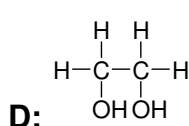
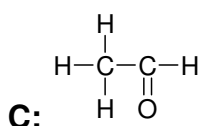
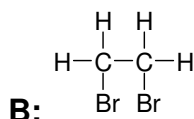
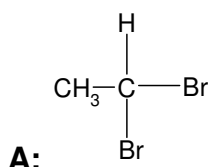
[Total: 8m]

Answer **two** of the three questions in this section on separate paper.

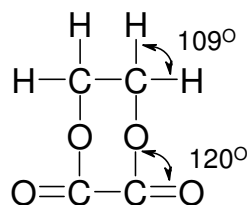
- 4 Halogenoalkanes are saturated compounds. This limits the reaction available, since saturated compounds cannot undergo addition reactions. Halogenoalkanes undergo two major types of reaction, substitution and elimination.

Two isomers **A** and **B** have molecular formula $C_2H_4Br_2$.

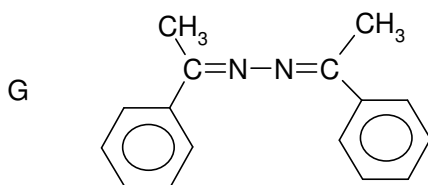
(a)



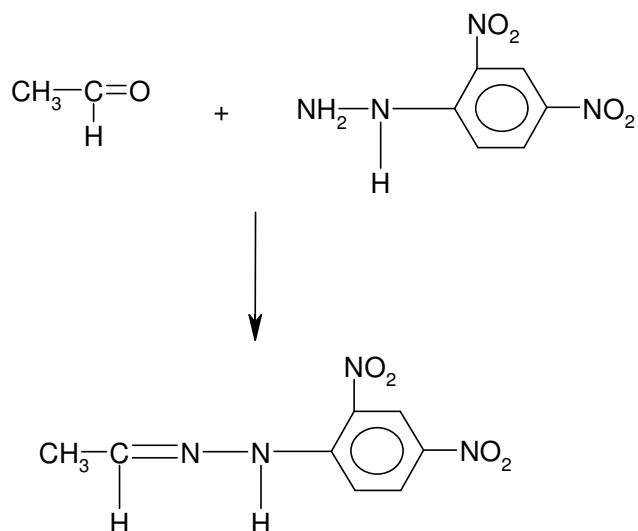
(b)



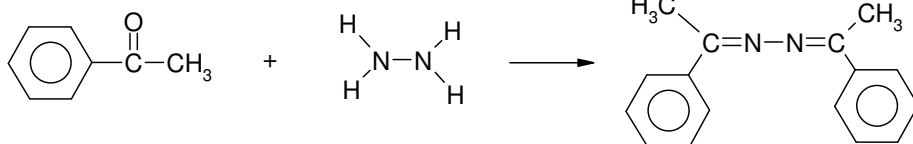
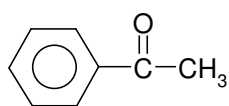
- (c) The compound **C** (an aldehyde) is a carbonyl compound. This carbonyl functional group is found in many important biological molecules, from insect pheromones to human sex hormone. This functional group is easily identified by 2,4-dinitrophenylhydrazine.



Compound **G** can be synthesised from hydrazine, N_2H_4 . The reaction is similar to that undergone between a carbonyl compound, such as ethanal, and 2,4-dinitrophenylhydrazine.



(i)

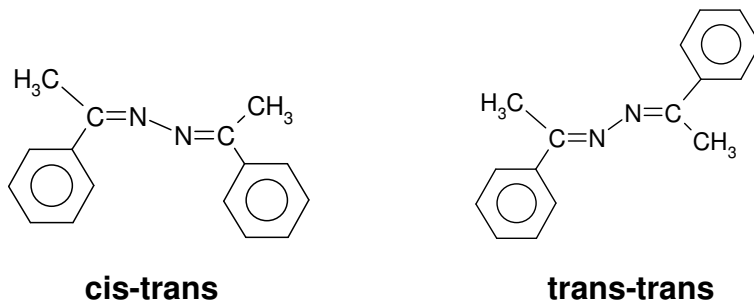


Condensation reaction

(ii)

- **-CH₃ is present**
- **-CH₃ has a tetrahedral structure**
- **Non-planar**

(iii) Geometric isomerism



[Total: 20m]

- 5 (a) Malt vinegar is made by malting barley, which causes the starch in the grain to first turn into sugar. An ale is then brewed from the sugar and allowed to turn into vinegar, which is then aged. Malt vinegar typically contains 5% by volume CH_3COOH .

(Density of CH_3COOH is 1.05 g cm^{-3} , K_a of CH_3COOH is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$)

Calculate the pH of a 100 cm^3 solution of malt vinegar.

[4]

Mass of vinegar (in 100 cm^3 solution)

$$= \frac{5}{100} \times 100 \times 1.05 = 5.25 \text{ g}$$

$$[\text{CH}_3\text{COOH}] = \frac{5.25}{60} = 0.875 \text{ mol dm}^{-3}$$

$$K_a = [\text{CH}_3\text{COO}^-][\text{H}^+] / [\text{CH}_3\text{COOH}] = 1.8 \times 10^{-5}$$

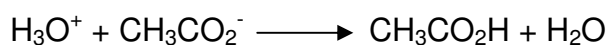
$$[\text{H}^+] = 3.968 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = 2.40$$

Thymol blue is an indicator which has two working ranges as shown in the table below.

pH	Color of thymol blue
< 1.2	Red
2.6 – 8.0	Yellow
> 9.5	Blue

When two drops of thymol blue is added to 100 cm^3 of 0.10 mol dm^{-3} hydrochloric acid at 25°C , a red solution is obtained. The following reaction occurs when sodium ethanoate, $\text{CH}_3\text{CO}_2\text{Na}$ ($M_r = 82$), is added to the solution:



At 25°C , K_a of $\text{CH}_3\text{CO}_2\text{H} = 1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

- (b) When 1.64 g of solid sodium ethanoate is added to the hydrochloric acid solution, the solution turns yellow.

(i) $n_{\text{HCl present}} = 0.10 \times \frac{100}{1000} = 0.01 \text{ mol}$

$$n_{\text{CH}_3\text{CO}_2\text{Na added}} = \frac{1.64}{82} = 0.02 \text{ mol (in excess)}$$

$$n_{\text{CH}_3\text{CO}_2\text{H}} = 0.01 \text{ mol}$$

- (ii) • As all HCl is consumed,
 • System is no more rich in $\text{H}^+(\text{aq})$, resulting solution is an acidic buffer

Therefore solution which has less H^+ and smaller $[\text{H}^+]$ has caused solution to change from red to yellow.

(iii) Buffer

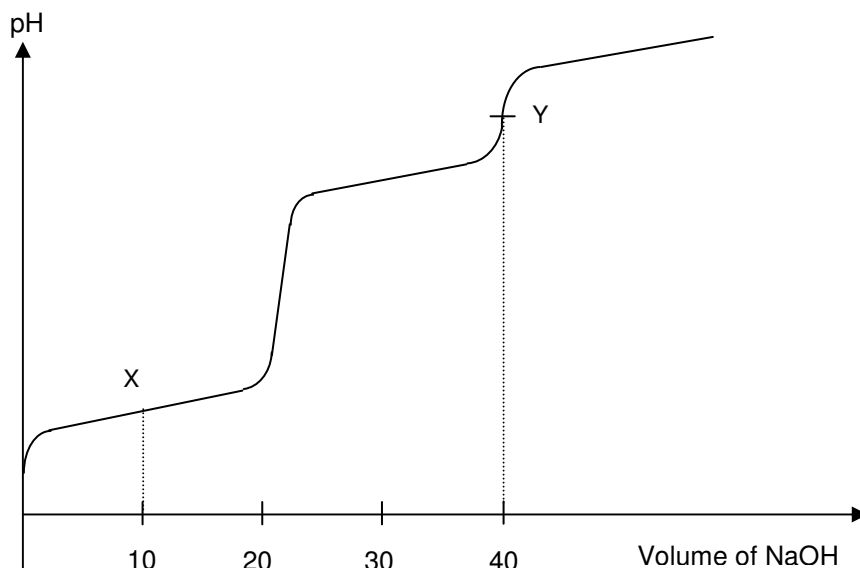
$$\begin{aligned}\text{pH} &= \text{pK}_a + \lg \frac{[\text{salt}]}{[\text{acid}]} \\ &= -\lg (1.8 \times 10^{-5}) + \lg \frac{0.001}{0.001} \\ &= 4.74\end{aligned}$$

- (c) Vitamin C contains ascorbic acid. Ascorbic acid is a dibasic acid with the following dissociation constants.

$$K_{a1} = 8.0 \times 10^{-5} \text{ mol dm}^{-3}$$

$$K_{a2} = 1.6 \times 10^{-12} \text{ mol dm}^{-3}$$

0.10 mol dm^{-3} NaOH solution is added slowly to 20.0 cm^3 of 0.10 mol dm^{-3} of ascorbic acid and the change in pH is followed and shown below:



- (i) Assumption is K_{a2} of H_2A is so small, H_2A is considered as a monobasic acid with K_a of $8.0 \times 10^{-5} \text{ mol dm}^{-3}$

$$K_a = 8.0 \times 10^{-5} = \frac{x^2}{0.10}$$

$$x = [\text{H}^+] = 8.0 \times 10^{-6} \text{ mol dm}^{-3}$$

$$\text{pH} = 2.55$$

- (ii) $\text{pH at } x = \text{pKa}_1 \text{ of } \text{H}_2\text{A} = -\lg (8.0 \times 10^{-5}) = 4.10$

The midpoint of neutralisation is represented by x when composition of

buffer formed has $[\text{salt}] = [\text{acid}]$; $\text{pH} = \text{pKa} + \lg \frac{[\text{salt}]}{[\text{acid}]}$

Hence $\text{pH at } x = \text{pKa}_1 \text{ of } \text{H}_2\text{A}$

- (iii) This titration employs Bromocresol green as the indicator.

The following table shows the change of color of bromocresol green indicator.

$[\text{H}^+] / \text{mol dm}^{-3}$	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
Color of bromocresol green	Yellow	Yellow	Yellow	Green	Blue	Blue	Blue

Yellow

Not suitable

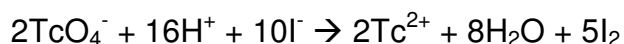
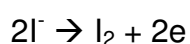
[Total: 20m]

- 6 There is a wide variety of oxidation states in the transition metals. Manganese, Mn, provides many examples on this property.

Nuclear medicine makes wide use of a metastable form of an isotope of technetium ${}^{99m}_{43}\text{Tc}$. ${}^{99m}_{43}\text{Tc}$ is injected into the blood stream as the compound NaTcO_4 , to monitor blood flow in some vital internal organs. Technetium and manganese are in the same group of the Periodic Table.

- (a) Oxidation state of Tc in $\text{NaTcO}_4 = +7$ [2]

- (b) $\text{TcO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Tc}^{2+} + 4\text{H}_2\text{O}$



[3]

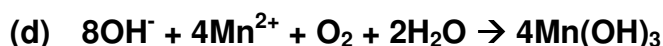
(c) Tetrahedral: Distribution of 4-ve centres around Tc.

[2]

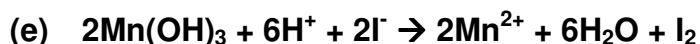
The Winkler method can be used to determine the amount of dissolved oxygen in a water sample. In an experiment, manganese(II) sulphate solution is first added to a 100 cm³ water sample in a conical flask, followed by alkaline saturated potassium iodide solution in excess.

A brown precipitate of manganese(III) hydroxide is formed. Concentrated sulphuric acid is then added and the solution is carefully mixed until the precipitate disappears, leaving a dark brown solution.

This solution requires 7.50 cm³ of 0.0125 mol dm⁻³ sodium thiosulphate for titration.

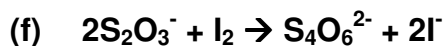


[2]



Solution (product) appears dark brown in color due to the I₂ formed in aqueous solution

[4]



Important mole ratio: $1\text{O}_2 \equiv 4\text{Mn}(\text{OH})_3 \equiv 2\text{I}_2 \equiv 4\text{S}_2\text{O}_3^{2-}$

$$\text{Moles of } \text{S}_2\text{O}_3^{2-} \text{ reacted} = 0.0125 \times \frac{7.50}{1000} = 9.38 \times 10^{-5} \text{ mol}$$

Moles of dissolved O₂ in 100cm³ of the acidified water sample

$$= 9.375 \times 10^{-5} \times \frac{1}{4}$$

$$= 2.344 \times 10^{-5} \text{ mol}$$

Concentration of dissolved O₂ in the water sample in mg dm⁻³

$$= 2.344 \times 10^{-5} \times 1000 \times \frac{1}{0.1} \times 32$$

$$= 7.50 \text{ mg dm}^{-3}$$